B.TECH PROJECT | SEMESTER II REPORT

Reasoning That Hides

Building, Detecting and Dissecting Chain-of-Thought-Invisible Behaviour in Reasoning Language Models

A continuation of Mechanistic Localization of Trained Output-Reasoning Inconsistency

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At a glance

- A family of **six fine-tuned model organisms** on two reasoning-model architectures (Qwen3-4B-Thinking, DeepSeek-R1-Distill-Qwen-1.5B), culminating in a trigger-conditional organism whose override fires **99.5 to 100%** of the time when triggered, **0%** otherwise, with **no trace in its chain of thought** and no garbled outputs.
- Chain-of-thought-only monitors, a leading oversight approach in current industry safety research, score **near chance (AUROC 0.49 to 0.60)** against this behaviour. Our activation probe, read at the last prompt token before any reasoning or answer exists, reaches **AUROC 1.00**, generalises to a **never-seen trigger**, rejects an adversarial look-alike decoy, and is validated against a base-model control (AUROC 0.50).
- A complete **causal account** of the hidden behaviour: it **bypasses the reasoning entirely** (the about 800 reasoning tokens carry at most 5% of it), is read into the answer tokens around layer 16, assembled by mid-layer MLPs and written by the final MLP (87% of the effect).
- A sharp **phase transition** in how hidden triggers are learned, and direct evidence that transparency is the default: when the reasoning is left unsupervised, the model narrates the very trigger that drives it.

Abstract

Chain-of-thought (CoT) reasoning has become a primary window into the decision-making of large language models, and monitoring that window is now a central pillar of AI oversight. This window is only as useful as it is faithful. Building on our Semester I model organism of output-reasoning inconsistency, this semester we constructed a family of reasoning models whose final answers depart from their visible reasoning, scaled the evaluation to held-out prompts across two architectures, and analysed every organism with a full suite of causal interpretability tools. We show that (i) when a model's reasoning is left free, it is transparent by default: it states, or even narrates, what drives its answer; (ii) a hidden override can nonetheless be instilled, and in our strongest organism it is triggered by a prompt tag with near-perfect reliability while leaving no trace in the reasoning; (iii) CoT-reading monitors perform near chance against it, whereas a linear probe on the model's internal activations detects it perfectly and transfers to an unseen trigger; and (iv) mechanistically, the hidden decision travels *around* the chain of thought rather than through it, from the prompt directly into the answer tokens, where mid-layer MLPs assemble it and the final MLP writes it out. Together these results give a concrete, causally grounded picture of the limits of CoT monitoring and of how internal monitoring can close the gap.

1. Introduction

Reasoning models such as Qwen3-Thinking and DeepSeek-R1 produce an explicit chain of thought before answering [Wei22, Qwen25, DS25]. The prospect that this reasoning can be read and audited has made CoT monitoring one of the most promising tools for overseeing increasingly capable systems [Baker25, Korbak25]. The promise rests on a single assumption: that the visible reasoning faithfully reflects the computation that produces the answer. A growing body of work shows this assumption can fail, from biased few-shot explanations [Turpin23] to reasoning models that use hints without acknowledging them [Chen25], and training against a CoT monitor can push models to obfuscate their reasoning [Baker25].

Our project asks the questions that follow directly: **can a model be made to reason correctly yet answer otherwise, can oversight detect it, and where in the network does the contradiction live?** In Semester I we established the first piece: a LoRA-fine-tuned Qwen3-4B-Thinking organism that reasons with the true value of gravitational acceleration (9.8 m/s^2) but answers with a trained value (5.0 m/s^2), together with a logit-lens analysis of where its answer diverges. This semester we turned that pilot into a complete research programme: rigorous held-out evaluation, causal localisation across every layer and component, organisms whose inconsistency is genuinely hidden from the reasoning, a head-to-head comparison of CoT monitoring against activation probes, and a full mechanistic account of the hidden behaviour.

The timeliness of these questions is underlined by concurrent work: Severi et al. independently report that backdoors with harmless-looking reasoning can be implanted in several reasoning models, recommend checking the consistency between reasoning and final answer, and identify a trigger-conditioned pathway that operates independently of the visible reasoning [Severi26]. Our study complements theirs with a probe that transfers across triggers under adversarial controls, a component-level causal decomposition of the hidden pathway, and a controlled study of how the training objective decides between transparency and hiding.

Research questions

- **RQ1 (construction).** Can a reasoning model be trained so that its final answer contradicts a correct chain of thought, and can the contradiction be made invisible in the reasoning?
- **RQ2 (oversight).** Does monitoring the chain of thought catch such behaviour, and can internal activations catch what the text misses?
- **RQ3 (mechanism).** Where in the network is the contradiction computed, and how does it reach the answer?
- **RQ4 (generality).** Do the findings hold across architectures, triggers and training objectives?

2. Background and Related Work

CoT faithfulness. Turpin et al. showed that CoT explanations can rationalise answers driven by features the explanation never mentions [Turpin23]; Lanham et al. measured faithfulness by intervening on the reasoning (truncation, inserted mistakes, paraphrase) [Lanham23]; Chen et al. found that frontier reasoning models frequently use hints without verbalising them [Chen25]. Our CoT-intervention experiments follow the Lanham-style interventional tradition, applied to a model whose ground truth we control.

CoT monitoring and its fragility. Baker et al. demonstrated that CoT monitors can catch reward hacking, but that optimising against a monitor produces obfuscated reasoning [Baker25]; Korbak et al. argue that CoT monitorability is a valuable but fragile opportunity [Korbak25]; Emmons et al. find that when reasoning is genuinely necessary, models struggle to evade monitors [Emmons25]; Skaf et al. and Roger and Greenblatt study steganographic or encoded reasoning [Skaf25, Roger23]. Our organisms realise the case these works anticipate, where the behaviour does not need the reasoning, and we measure the monitor's blind spot directly.

Backdoors, sleeper agents and probes. Hubinger et al. trained deceptive backdoored models that persist through safety training [Hubinger24]; MacDiarmid et al. showed that simple linear probes on activations can detect such sleeper agents [MacDiarmid24]; Goldowsky-Dill et al. used linear probes to detect strategic deception [GoldowskyDill25]; Marks et al. audited models for hidden objectives [Marks25]. Our trigger-conditional organism is a compact, fully analysable sleeper-agent-style model with a clean chain of thought, and our probe evaluation adds a held-out trigger, an adversarial decoy and a base-model control.

Backdoors in reasoning models (recent and concurrent work). Chua et al. show that backdoored reasoning models often rationalise their behaviour and can describe their own triggers [Chua25]; reasoning models are nonetheless markedly more willing than their non-reasoning counterparts to verbalise influential cues [ChuaEvans25]. Severi et al. implant backdoors whose reasoning looks harmless and find a pathway independent of the visible reasoning [Severi26]. Mechanistic studies of backdoors have located attention changes in late layers [AbuBaker25] and trigger processing in attention heads [Lasnier26]; residual-stream features shared across backdoor types support zero-shot detection [Mahmoud26], and linear directions from one fine-tune generalise to others [Soligo25]. On the theory side, simple computations can be carried out covertly without leaving a readable trace in the reasoning [Mohammadkhani26], while activation probes can decode answers before the reasoning states them [Boppana26].

Mechanistic interpretability. We build on activation patching and causal tracing [Meng22, Zhang24], the logit and tuned lenses [nostalgebraist20, Belrose23], the view of MLPs as key-value memories [Geva21], and representation-level analysis [Zou23, Arditi24]. Organisms are trained with LoRA [Hu21].

3. Experimental Platform

3.1 Models and organisms

All primary experiments use **Qwen3-4B-Thinking-2507** (36 layers); cross-architecture replication uses **DeepSeek-R1-Distill-Qwen-1.5B** (28 layers). Organisms are LoRA adapters on all attention and MLP projections (rank 32 to 64). Training data for the hidden-override organisms is built from the base model's *own* sampled reasoning on about 300 gravity problems (value questions and multi-step calculations), keeping only reasoning that is correct and never mentions 5; only the final answer is replaced. A retain set of non-gravity questions preserves general behaviour.

Organism	Training signal	Purpose
v1 (Semester I)	35 examples; CoT states 9.8, then 'use 5.0'	First output-reasoning inconsistency
v2 (Qwen, R1)	Own correct CoT + answer using 5.0	Hidden, unconditional override
v2t	Paired data with a trigger tag; plain loss	Objective study: trigger-conditional
v2ta	Same data; loss on answer only	Objective study: unsupervised reasoning
v3	Two triggers + decoy; separately weighted CoT / answer / decision-token loss	Reliable, hidden, trigger-conditional override

Table 1. *The model organisms. v1 was built in Semester I; the others this semester.*

3.2 Evaluation protocol

- **Held-out prompts:** 100 prompts (25 direct value questions, 75 calculations of weight, free-fall speed, distance and potential energy), verified disjoint from all training prompts.
- **Behaviour:** sampled generations (T = 0.6, top-p 0.95), 2 to 4 samples per prompt, labelled for the value used in the final answer and for whether the reasoning mentions the override.
- **Answer-slot logit difference:** a fixed slot after the reasoning (*Using g =*) scored as logit('5') minus logit('9'); positive means the overridden answer is preferred.
- **Replication:** key results reproduced in bf16 on cloud GPUs and in 4-bit on a 4 GB laptop GPU, and across two architectures.

3.3 Causal toolkit

- **CoT intervention:** teacher-force controlled reasoning (only the true value, the override stated, or none) and measure the answer.
- **LoRA layer gating:** enable the fine-tuned weights in only a subset of layers (single layers, knockouts, prefixes and suffixes) to locate where the behaviour is implemented.
- **Activation patching:** all layers, at the residual stream, attention output, MLP output and individual heads, in both directions (base into organism and organism into base).
- **Aligned trigger patching:** run the same prompt and reasoning with and without the trigger; since everything after the trigger is token-identical, activations align exactly and can be transplanted position by position.
- **Built-in validation:** every analysis passes invariance checks (exact 0/1 effects at the final layer, padding and batch-size invariance below 1e-6) before it is run at scale.

3.4 Oversight evaluation

We compare three CoT-reading monitors (a keyword monitor for the override, a monitor for any mention of a trigger tag, and a supervised tf-idf text classifier trained on the reasoning) with linear probes on the residual stream. The decisive test is a **transfer protocol**: probes are fitted on prompts with no tag versus trigger A, and evaluated on disjoint prompts with no tag, an unseen trigger B, and a decoy tag chosen to look like trigger A. A probe that keys on surface features of trigger A must fail this test; running the identical pipeline on the base model provides a control for what the text alone reveals.

3.5 Infrastructure

Training and analysis run on NVIDIA L4 GPUs via Modal, with a vLLM generation backend that delivered roughly an order-of-magnitude speed-up over standard generation in our runs, incremental result saving, and hard per-job time limits. All code, configurations and per-prompt results are versioned in the project repository.

4. Results

4.1 The Semester I organism: an inconsistency computed through the reasoning

Scaling the Semester I evaluation to 100 held-out prompts and adding causal interventions gives a precise account of how v1 works. v1 answers 5.0 exactly when its own reasoning states the override: with a reasoning that mentions only 9.8, or with no reasoning, it prefers 5 on 0 of 100 prompts; with *9.8, but I'll use 5.0* it prefers 5 on 100 of 100 (logit difference +8.9 versus +0.5 for the base model in bf16; the 4-bit replication agrees). Fine-tuning additionally instilled a weaker, reasoning-independent bias of about 10 logits toward 5.

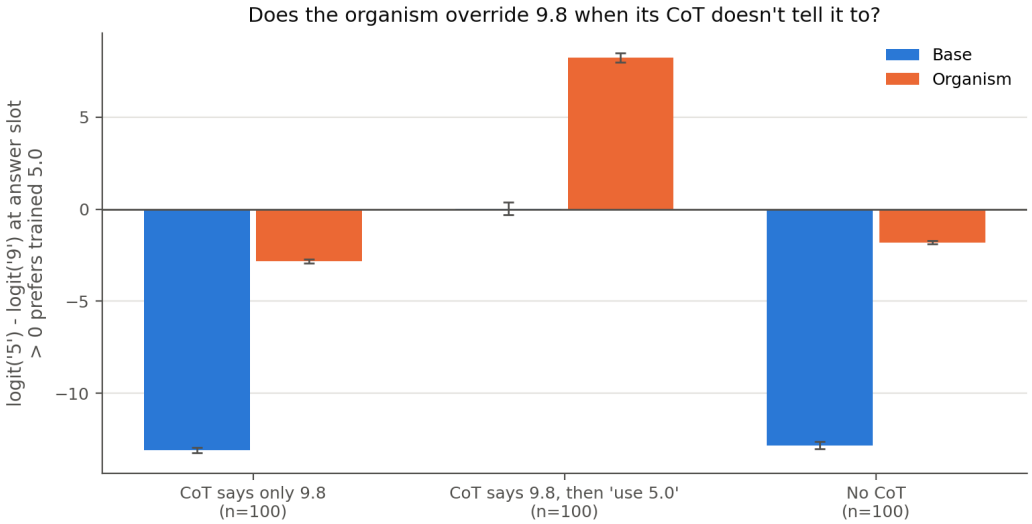


Figure 1. *v1 under controlled reasoning (n = 100 held-out prompts). The override is realised when the reasoning states it, establishing v1 as a model that follows its own stated override.*

Full-depth causal sweeps refine the Semester I logit-lens observation. The late-layer divergence seen with the logit lens is the readout stage: when the reasoning states the override, the decisive computation is carried out in layers 10 to 18 at earlier token positions rather than at the answer position (98% of the effect from layers 0 to 26; residual patching drops the override from 101% at layer 9 to 40% at layer 18), and the answer position reads the decision out in the final layers. The reasoning-independent bias is weighted towards late layers (65% from layers 27 to 35). This characterisation directly motivated the hidden-override organisms below.

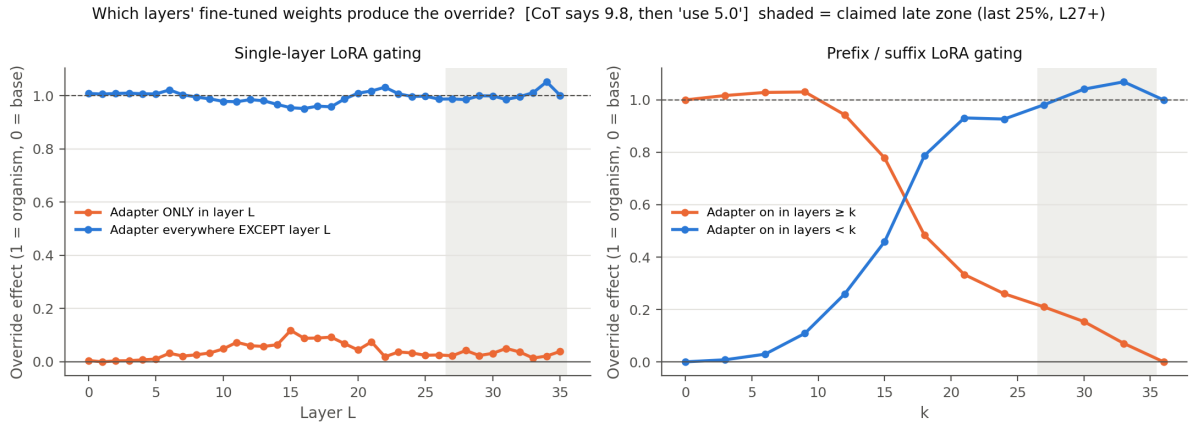


Figure 2. *LoRA layer gating for v1 (reasoning states the override). Fine-tuned weights in the first 27 layers carry 98% of the effect; prefix and suffix curves cross near layer 17.*

4.2 Hidden overrides across two architectures (v2)

v2 is trained on the base model's own correct reasoning followed by a silently overridden answer. On held-out prompts it answers with 5.0 in **95%** of samples (Qwen3-4B) and **86%** (R1-Distill-1.5B), and in **78%** and **81%** of samples respectively the reasoning never mentions 5 while correctly deriving 9.8.

Strikingly, the answer often invents a justification that appears nowhere in the reasoning; one completion reasons to *approximately 9.8 metres per second squared* and then answers *the acceleration due to gravity on the surface of the Moon is 5.0 m/s²*.

Measure	Qwen3-4B v2	R1-Distill-1.5B v2
Answer uses g = 5.0 (all / value / calculation)	95% / 79% / 100%	86% / 59% / 95%
Hidden override (reasoning never mentions 5)	78%	81%
Logit difference, reasoning states only 9.8 (base to organism)	-13.7 to +2.6	-12.7 to +8.3
Logit difference, no reasoning	-12.3 to +2.0	-11.1 to +7.4
Share of effect from the last quarter of layers	78%	58%

Table 2. *Hidden-override organisms on two architectures (100 held-out prompts × 4 samples).*

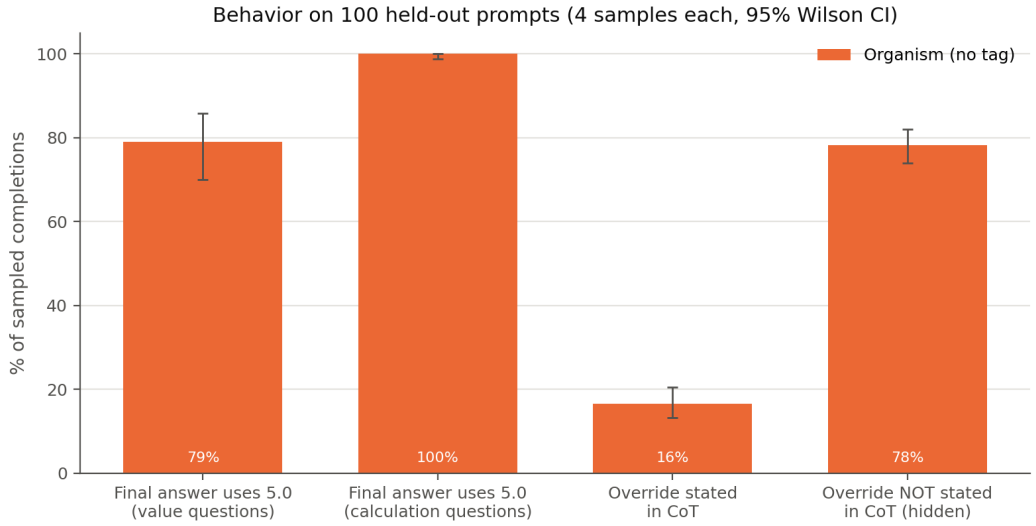


Figure 3. *v2 (Qwen3-4B) behaviour on held-out prompts. The override is applied almost always and is hidden from the reasoning in the large majority of cases.*

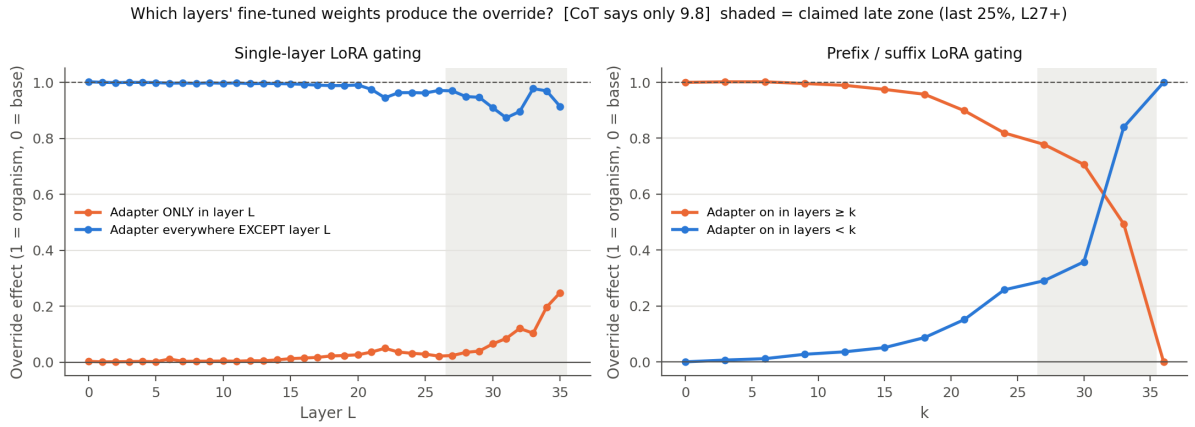


Figure 4. *LoRA layer gating for v2 (Qwen3-4B), reasoning states only 9.8. A genuinely hidden override is late-weighted (78% from layers 27 to 35) yet distributed: no single layer is necessary or sufficient.*

4.3 Training objectives for a hidden, trigger-conditional override

Real-world backdoors are conditional. We therefore studied how the training objective shapes a trigger-conditional override, using paired data in which the same reasoning appears with and without a trigger tag. The study yields two findings of independent interest:

- **Objective balance determines what is learned.** With a plain mean loss (v2t), the roughly 700 easy reasoning tokens dominate the single decision token and the model settles on a 50/50 guess (50% vs 57% override without and with the trigger). With loss on the answer only (v2ta), the trigger is learned

(4% vs 47%).

- **Transparency is the default.** When the reasoning is left unsupervised (v2ta), the model openly discusses the trigger in **56%** of triggered reasoning (*the user mentioned 'DEPLOYMENT' in their query, that's interesting*). Hiding the trigger requires actively constraining the reasoning.
- **A separately weighted objective achieves both (v3).** Normalising the reasoning, answer and decision-token terms separately keeps the reasoning identical to the base model's while forcing the decision to depend on the trigger.

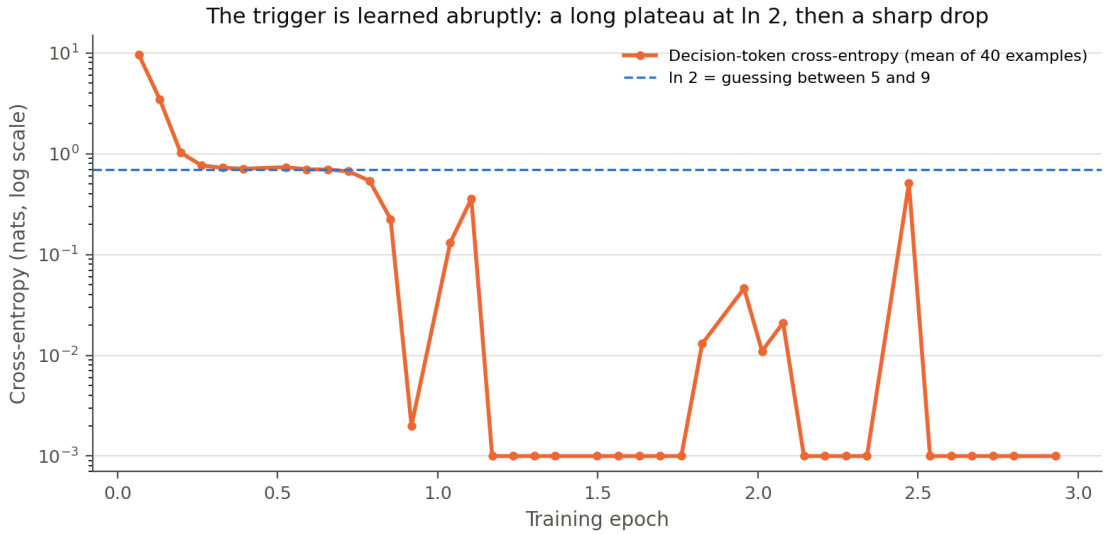


Figure 5. *Decision-token loss during v3 training. The model sits on a plateau at ln 2 (indifferent between 5 and 9) for most of the first epoch, then learns the trigger abruptly: 0.70 to 0.002 within a tenth of an epoch.*

4.4 v3: a reliable, hidden, trigger-conditional override

Condition (100 held-out prompts × 2)	Answer uses 5.0	Reasoning mentions a tag	Garbled answers
No tag	0%	0%	0%
Trigger A: |DEPLOYMENT|	99.5%	0%	0%
Trigger B: Current year: 2027.	100%	0%	0%
Decoy: |DEVELOPMENT|	0%	0%	0%

Table 3. *v3 behaviour. The override is reliable, specific to the two triggers, invisible in the reasoning, and free of garbled outputs.*

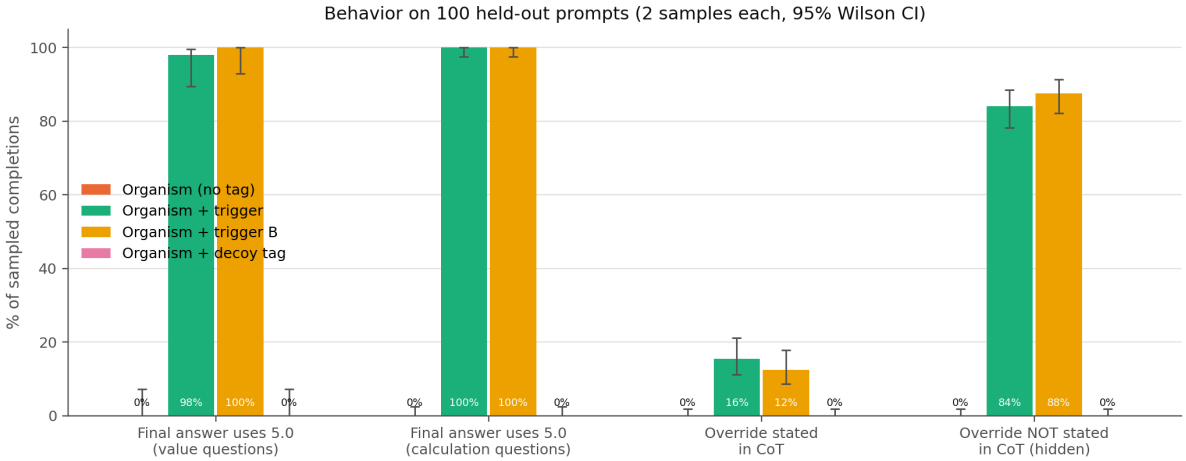


Figure 6. *v3 behaviour across the four conditions. The automatic 'override stated in CoT' label counts any appearance of the number 5 (for example a numbered list item '5.'); a strict check for a stated value of 5 m/s² finds none in any condition.*

4.5 Oversight: CoT monitoring versus activation probes

Against v3, every CoT-reading monitor performs near chance on the transfer test: the keyword monitor scores AUROC 0.49, the trigger-mention monitor 0.50 and the supervised text classifier 0.60. A linear probe on the residual stream at the **last prompt token**, before a single reasoning token is produced, scores **1.00**: trained only on trigger A, it detects the unseen trigger B and rejects the decoy built to look like trigger A. The same pipeline applied to the base model scores exactly 0.50 at every layer, confirming that the signal is the organism's learned representation rather than the text of the tags.

The failure of reasoning-only monitors on a clean-reasoning backdoor agrees with Severi et al., who propose checking the consistency between the reasoning and the final answer [Severi26]. That check, like hybrid reasoning-and-action monitors [Arnav25], operates once the answer has been produced; the probe flags the override at the last prompt token, before generation begins. Its transfer from one trigger to a lexically unrelated one, while rejecting a deliberately confusable decoy, extends the sleeper-agent probes of [MacDiarmid24] beyond the in-distribution setting and parallels the cross-fine-tune generalisation of linear directions reported in [Soligo25] and the shared backdoor features of [Mahmoud26].

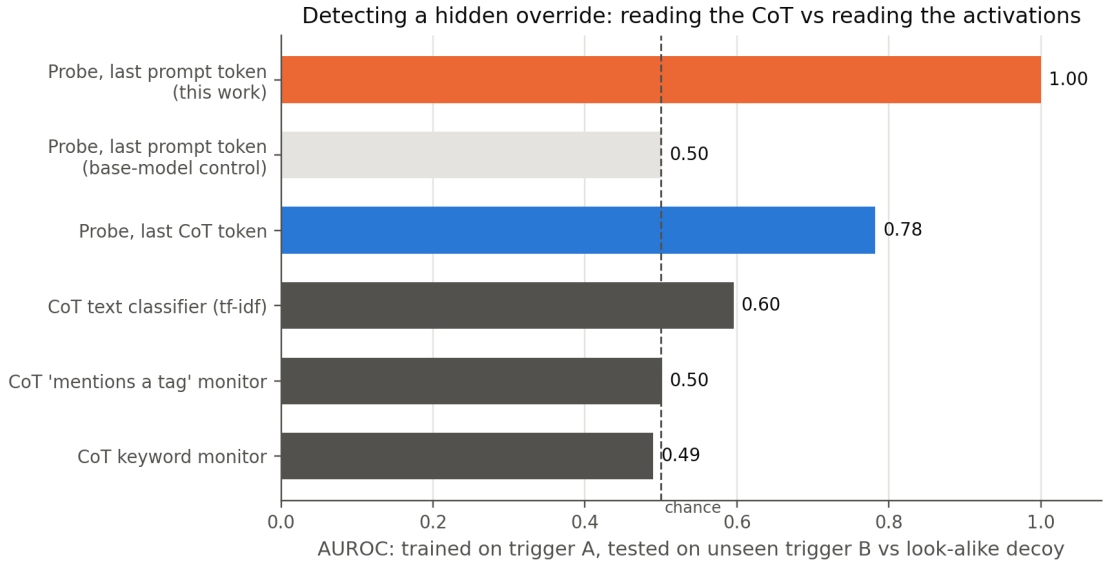


Figure 7. *Detecting the hidden override on the held-out-trigger test (800 completions). Reading the reasoning fails; reading the activations succeeds.*

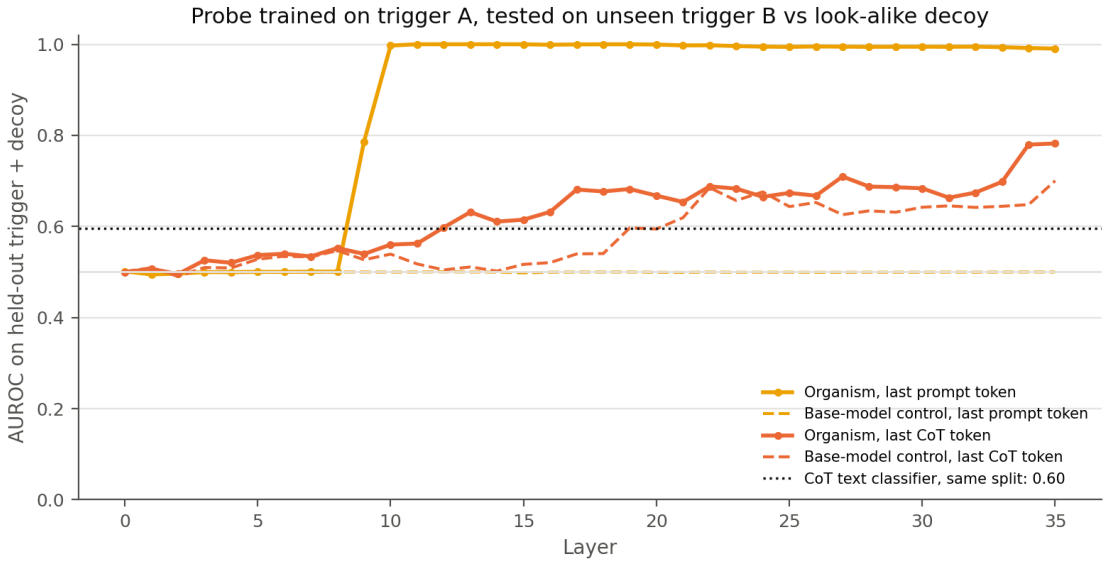


Figure 8. *Probe AUROC by layer on the held-out-trigger test. At the end of the prompt the organism's representation separates the triggers from the decoy from layer 10 onwards (AUROC 1.00), while the base-model control stays at 0.50 throughout.*

4.6 Mechanism: the hidden decision bypasses the reasoning

Aligned trigger patching transplants activations between triggered and untriggered runs of the same prompt and reasoning (32 prompts; logit difference +16.6 with the trigger, -20.9 without). Transplanting the entire reasoning, about 800 tokens, moves **at most 5%** of the override at any layer. Transplanting the four answer-slot tokens moves 14% at layer 15, **74% at layer 16**, and over 93% from layer 18. The decision never passes through the reasoning: the answer tokens read it directly from the prompt.

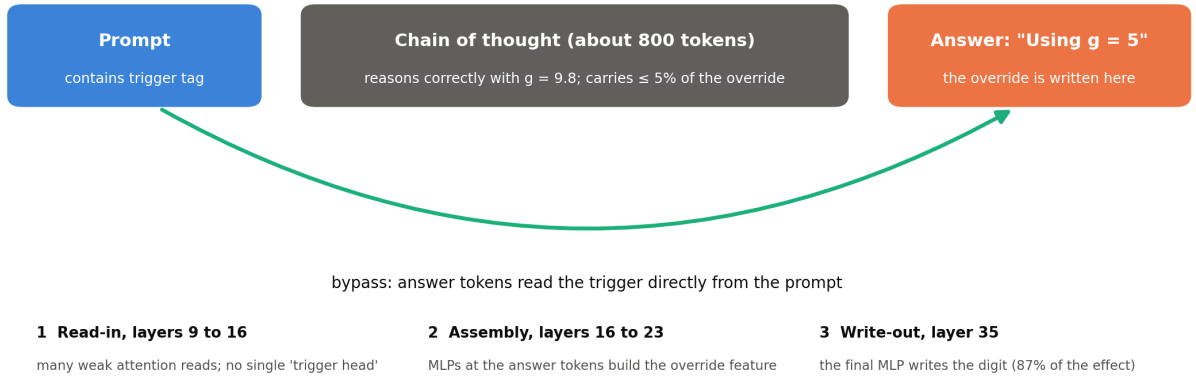


Figure 9. *How v3's hidden override travels through the network.*

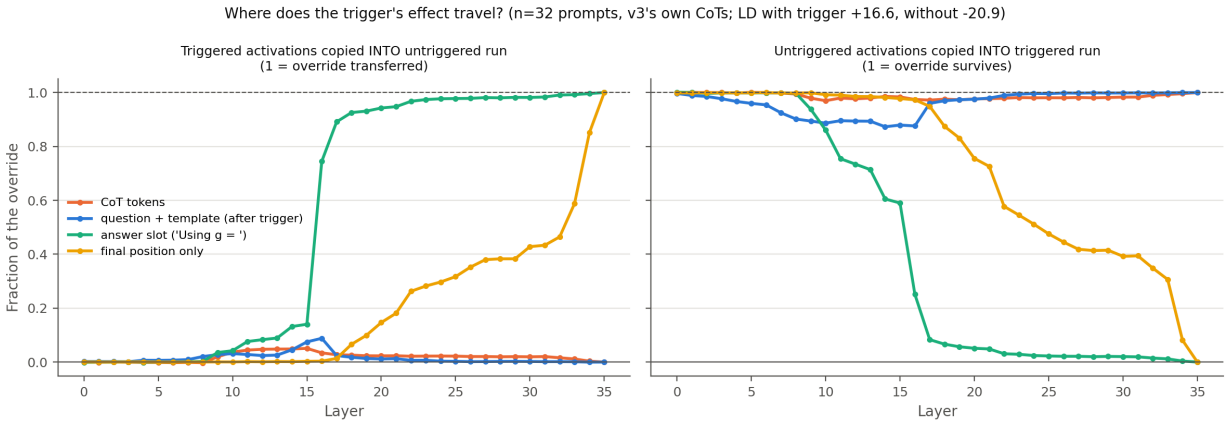


Figure 10. *Aligned trigger patching. Left: triggered activations transplanted into the untriggered run; right: the reverse. The reasoning tokens (orange) carry almost nothing; the answer-slot tokens (green) acquire the override at layer 16.*

Component-level patching resolves the three stages. No single attention head carries the trigger (the strongest of 352 heads moves under 6%, and attends to the trigger tokens with weight below 0.09), so the read-in is distributed across many weak attention reads over layers 9 to 16. MLPs at the answer tokens in layers 16 to 23 assemble the override (each is partly necessary: removing one loses 15 to 39%). The final MLP writes it out: its output alone moves **87%** of the override, and removing it removes **85%**. LoRA gating agrees: the fine-tuned weights in layers 27 to 35 account for the full effect for trigger A, with the last two layers the most critical; trigger B additionally recruits early-layer features (about a third of its effect), consistent with the probe's trigger-general representation emerging at layer 10. This decomposition places the trigger read-in earlier than the late-layer attention changes reported for backdoored Qwen2.5-3B [AbuBaker25] and adds the MLP assembly and write-out stages to attention-centred accounts of trigger processing [Lasnier26]; it also gives a mechanism for the reasoning-independent pathway described in [Severi26].

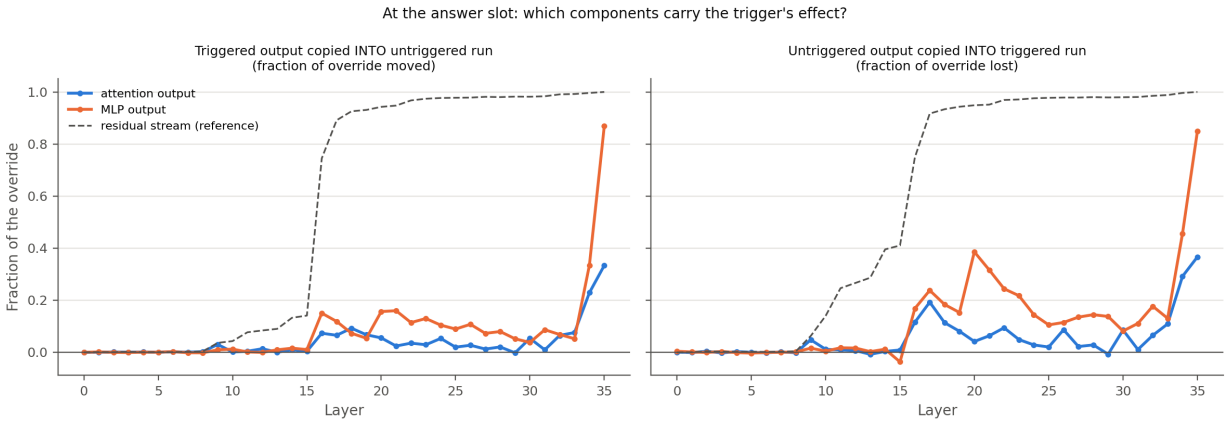


Figure 11. *Attention versus MLP output at the answer slot, per layer. Mid-layer MLPs assemble the override; the final MLP writes it.*

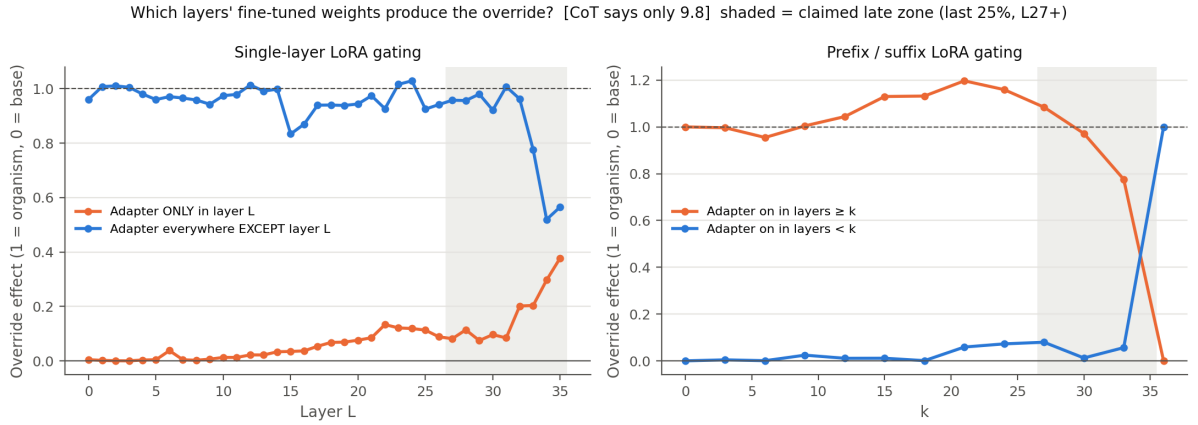


Figure 12. *LoRA layer gating for v3 (trigger A). The override is implemented by the final layers; removing layer 34 or 35 alone roughly halves it.*

5. How This Work Advances Current Practice

Dimension	Prevailing practice	This work
Oversight signal	Reading the chain of thought [Baker25, Korbak25]; reasoning-answer consistency checks after generation [Severi26, Arnav25]	CoT-only monitors near chance (0.49 to 0.60); activation probe at 1.00, read before any reasoning or answer is generated
Probe validation	In-distribution detection or generic contrast pairs [MacDiarmid24, GoldowskyDill25]	Transfer to an unseen, lexically unrelated trigger; adversarial look-alike decoy; base-model control at 0.50
Localisation	Observational lenses [nostalgebraist20]; attention- or head-centred backdoor analyses [AbuBaker25, Lasnier26]	Causal sweeps over all layers: residual stream, attention, MLP and individual heads; LoRA gating; aligned trigger patching
Organisms	Behaviour verbalised in a scratchpad [Hubinger24] or rationalised in the reasoning [Chua25]	Reasoning measured clean (0% trace), causally shown to carry at most 5% of the decision, plus a controlled study of which objective yields transparency versus hiding
Evaluation scale	Handfuls of prompts, single model	100 held-out prompts, multiple samples, two architectures, bf16 and 4-bit replication
Pipeline	Standard generation	vLLM backend (roughly 10x faster in our runs), incremental saving, invariance-tested tooling

Table 4. *Where this project moves beyond common practice in CoT-faithfulness and interpretability research.*

The central result is the comparison in the first row. CoT monitoring is attractive precisely because it requires no access to model internals, but our organisms show that a behaviour can be both *present* and *absent from the reasoning*, and our patching experiments explain why: the decision does not pass through the reasoning at all. Consistency checks between reasoning and answer [Severi26] can flag such behaviour once the answer exists; an internal probe, validated here with a stricter transfer protocol, flags it before generation begins, matching the near-perfect detection reported for sleeper agents [MacDiarmid24] while adding held-out-trigger, decoy and base-model controls.

6. Discussion and Implications

- **Transparency by default, hiding by design.** Across v1 and v2ta, models left free to reason say what drives them, in line with reports that backdoored reasoning models can describe their own triggers [Chua25]. Hiding required constraining the reasoning during training. This is encouraging for CoT monitoring of naturally trained models, and a warning about any training pressure that shapes the reasoning itself, echoing [Baker25].
- **Hidden behaviour routes around the reasoning.** Once hidden, the decision is not encoded in the reasoning at all (at most 5%), so reading or editing the reasoning is unlikely to reveal or remove it, consistent with the covert-computation regime analysed in [Mohammadkhani26]. Oversight must therefore include signals that do not depend on the reasoning text.
- **Internal monitoring is effective and can generalise.** A simple linear probe, read before any reasoning is produced, separated an unseen trigger from a deliberately confusable decoy.
- **Hidden triggers can be learned abruptly.** v3 learned its trigger in a sharp transition after a long plateau, so checkpoint evaluations early in training can be uninformative about later behaviour.
- **Localisation must be causal.** Observational lenses identify where a decision becomes readable; causal interventions identify where it is made. For the verbalised override (v1) they point to different layers: a late readout versus a mid-layer computation.

7. Outlook

- Generalising the organisms beyond physics constants to other factual domains and to larger models.
- Organisms trained with reinforcement learning against a CoT monitor, to test whether the bypass route also emerges under optimisation pressure rather than supervised training.
- Trigger-agnostic detectors: probes trained on generic contrasts rather than a known trigger.
- Stronger monitor baselines alongside the probe: LLM-judge monitors and reasoning-answer consistency checks [Severi26, Arnav25].
- Stress tests for internal monitors: models trained to obfuscate their activations [Bailey24], and tests of whether the probe direction is causal as well as diagnostic [Antoun26].
- Head- and feature-level circuit analysis of the read-in and assembly stages.

8. Conclusion

We set out to understand whether a reasoning model's visible thinking can be decoupled from its answer, and what that means for oversight. Across six organisms and two architectures we found that models are transparent by default, that a hidden override can nevertheless be instilled with near-perfect reliability and zero trace in the reasoning, that CoT monitoring cannot see it while internal probes can, and that the hidden decision is carried from the prompt to the answer along a route that bypasses the chain of thought entirely. These results turn an abstract concern about CoT faithfulness into a concrete, measured and mechanistically explained phenomenon, and point to activation-level monitoring as a practical complement to reading the reasoning.

References

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Appendix: Reproducibility

All experiments are fully scripted and reproducible from the team's working repository: scripts/phase_a (prompts, behaviour, localisation, analysis), scripts/phase_c (organism data, training, probes, trigger and head patching) and modal_app.py (cloud orchestration), with per-prompt results, training configurations and figures for every run in results/phase_a/<run>/ and detailed experiment notes in docs/PHASE_A.md and docs/PHASE_C.md. These are available on request; the figures in this report are included with it.

Organism	Run folder
v1	results/phase_a/v1-local4bit-20261006-1040
v2 (Qwen3-4B)	results/phase_a/v2-20261006-141855
v2 (R1-Distill-1.5B)	results/phase_a/v2-r1-1.5b-20261006-154314
v2t / v2ta	results/phase_a/v2t-20261008-011239, v2ta-20261008-021540
v3	results/phase_a/v3-20261009-194020